

ORF 245: Fundamentals of Statistics – Spring 2025

Precept 9

1. Linear Regression and Least Squares

(a) Statistical model

$$y = \beta_1 x + \beta_0 + \varepsilon$$

- Observe: x **explanatory variable**; y **response variable**.
- Not observed: ε **noise** (what the explanatory variable can not explain).
- Model parameters: β_1 and β_0 .
- Target: to estimate β_1 and β_0 using i.i.d. observations $(x_1, y_1), \dots, (x_n, y_n)$ sampled from the above model.
 - One critical assumption: **exogeneity** $\mathbb{E}[\varepsilon|x] = 0$.

(b) Linear Least Squares

- Optimization problem $(\hat{\beta}_1, \hat{\beta}_0) = \operatorname{argmin}_{\beta_1, \beta_0} \sum_{i=1}^n (y_i - \beta_1 x_i - \beta_0)^2$ (Training).
 - A geometric interpretation: find a line such that the total sum of the square of the difference between fitted value $\hat{y} = \hat{\beta}_1 x + \hat{\beta}_0$ and observed value y are smallest.
 - Optimal solution (solution is unique if there are distinct x_i 's)

$$\hat{\beta}_1 = \frac{s_{x,y}}{s_{x,x}} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \bar{x}\hat{\beta}_1$$

$$\text{where } \bar{a} = \frac{1}{n} \sum_{i=1}^n a_i \text{ and } s_{a,b} = \sum_{i=1}^n (a_i - \bar{a})(b_i - \bar{b}).$$

- Fitted model
 - **Fitted values** $\hat{y}_i = \hat{\beta}_1 x_i + \hat{\beta}_0$, **residuals** $\hat{\varepsilon}_i = y_i - \hat{y}_i$.
 - **Error sum of squares (SSE)** and **total sum of squares (SST)**

$$\text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad \text{and} \quad \text{SST} = \sum_{i=1}^n (y_i - \bar{y})^2$$

* Interpretation of SST: variations of the response variable y ($\frac{1}{n-1} \text{SST} \approx$ its variance);

* Interpretation of SSE: the variations in y that the linear model does not explain.

• One can further use SSE to estimate $\sigma^2 = \mathbb{E}[\varepsilon^2]$.

• **Residual standard error** $\hat{\sigma} = \sqrt{\frac{\text{SSE}}{n-2}}$.

* These are absolute values.

- **coefficient of determination** / R^2

$$r^2 = 1 - \frac{\text{SSE}}{\text{SST}}$$

* Interpretation of R^2 : how much proportion of y variations can be explained by linear model.

* $R^2 = 0 \rightsquigarrow \hat{\beta}_1 = 0$, $R^2 = 1 \rightsquigarrow y_i = \hat{\beta}_1 x_i + \hat{\beta}_0$.

- Prediction and Interpretation

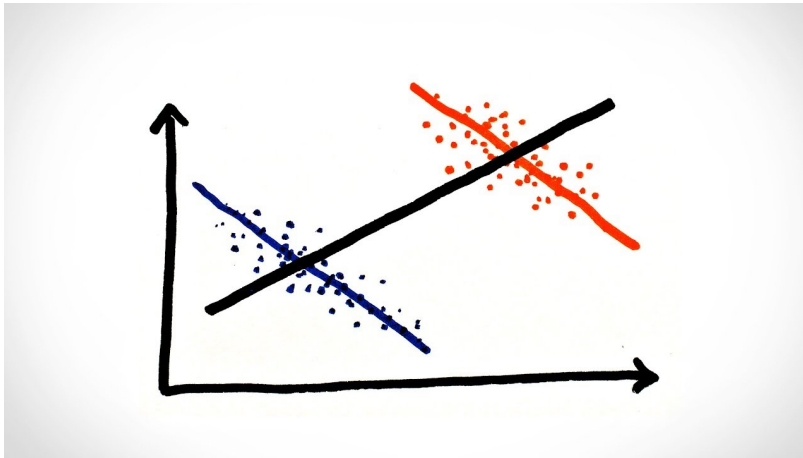
* Given some new input x_{test} , use $y_{\text{test}} = \hat{\beta}_1 x_{\text{test}} + \hat{\beta}_0$ to predict the response variable.

- * Interpretation of β_1 : One can observe the expectation of y goes up by β_1 if when x goes up by one unit in the observational data.
 - This is the association based on observed data, not causality.
 - For causality, we hope if we get involved, increasing one unit of x will lead to a β_1 increase in y .

2. **(Interpretation of linear model in the real world)** A quantitative researcher wants to get some preliminary understanding of two variables x and y using observations and linear least squares. He/She first tries to regress y on x using data in the whole day and gets a significant positive $\hat{\beta}_{1,full}$ (which means $\hat{\beta}_{1,full} > 0$ and SSE is significantly smaller than SST). Then he/she tries to regress y on x using data in the morning/noon/afternoon respectively and gets a significantly negative $\hat{\beta}_{1,m}$, $\hat{\beta}_{1,n}$ and $\hat{\beta}_{1,a}$.

[Q]: Is it possible that the above scenario will happen? If the answer is no, please provide formal proof. If the answer is yes, please explain the intuitions (for example, a geometric illustration or specify a concrete example) and which association (negative/positive) he/she can use in the real world.

- **Solution:** Yes, it is possible that the above circumstance will happen. See the following illustration, as an example when data is split into two (instead of three) groups:



Whether to use the $\hat{\beta}_1$ calculated in sub-groups depends on the true causality structure.

We consider another real-world example (page 3 and Figure 1.1 in (Pearl et al. 2016)) to illustrate.

Model 1

- The response variable y is the cholesterol, and the explainable variable x is the weekly exercise.
- We collect data from people ages ranging from 10 to 69 and run linear regression, the linear model returns a significantly positive $\hat{\beta}_1$.
- If we divide the whole dataset into three sub-groups, young (10-29), mid ages (30-59), old (50-69), and run least squares on the three sub-groups separately, we all get negative β_1 coefficients.
- This shares a similar spirit with that of the scenario above.

Why is it the case?

- The causal model (how data are generated).
 - * For each sub-group, the coefficient matches our understanding of causality (more exercise you did $\rightsquigarrow$ lower cholesterol).

- * For the whole population, older people tend to have higher cholesterol and also tend to do exercise (more spare time, need to do exercise to keep healthy).
- Prediction based on the explainable variable x and the whole dataset. (how fitted model works)
 - * The least squares aim to fit a model to explain the data better.
 - * So it implicitly does the following procedure: more exercise time $\rightsquigarrow$ more likely to be an older person $\rightsquigarrow$ higher cholesterol.
- This is a continuous version of the Simpson's Paradox we discussed before.
- Though the first whole dataset model is good for prediction, it will lead to false scientific claims and cannot guide us to make decisions in the real world. (difference between association and causality)

Model 2: Does grouping always help?

- Focused on 30 years-old people, x weekly exercise in 2022, y cholesterol measured in the first week of 2023, consider the three groups: health score measured at the same time (group 1: 80-100, group 2: 60-79, group 3: 0-59).
- if we regress y on x on the whole dataset, then $\hat{\beta}_1 < 0$ is a causal relationship.
- if we regress y on x on each sub-groups, $\hat{\beta}_1$ may be positive. Consider the following logic: more exercise is a direct cause of a decrease in cholesterol, and lower cholesterol is a direct cause of a higher health score, while more exercise will also (positively) affect the health scores in manners other than cholesterol. Then, for a fixed health score, more exercise implies that other measurements in the health score will be better, and thus implies that the cholesterol measurement is worse.

Comparison and take-home message

- In model 1, one should use $\hat{\beta}_1$ in sub-groups to guide our decision. In model 2, one should use $\hat{\beta}_1$ in the whole dataset to guide our decision.
- This is because the causality structures behind the two models are different. Let z be the variable concerning the sub-group id. For the first model, z is a direct cause of x and y , x is the direct cause of y ; for the second model, x is the direct cause of y and z , y is the direct cause of z .
- For the same model but with different causal structures, one should use different adjustments to obtain a correct scientific conclusion.

3. **(Simple Linear Regression, Exercise)** The sales of a company (in million dollars) for each year are shown in the table below.

x (year)	2005	2006	2007	2008	2009
y (sales)	12	19	29	37	45

- Find the least square regression line $y = \hat{\beta}_0 + \hat{\beta}_1 x$.
- Use the least squares regression line as a model to estimate the sales of the company in 2012.

• **Solution:**

- We first change the variable x into t such that $t = x - 2005$ and therefore t represents the number of years after 2005. Using t instead of x makes the numbers smaller and therefore manageable. The table of values becomes.

t (years after 2005)	0	1	2	3	4
y (sales)	12	19	29	37	45

We now calculate $\hat{\beta}_0$ and $\hat{\beta}_1$ using the least square regression formulas (See details in notes):

$$\hat{\beta}_1 = 8.4$$

$$\hat{\beta}_0 = 11.6$$

– In 2012, $t = 2012 - 2005 = 7$.

The estimated sales in 2012 are:

$$y = 8.4 \times 7 + 11.6 = 70.4 \text{million dollars}$$